

Math 526, S16

Quiz 6

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ (the reduced echelon form) to find the condition on b_1, b_2, b_3 such that $Ax = b$ is solvable.

$$\begin{aligned}
 [A \ b] &= \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 2 & 6 & 4 & 8 & b_2 \\ -2 & -6 & 0 & 0 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 + 2b_1 \end{array} \right] \\
 &= \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] \\
 &= \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 1 & 2 & \frac{b_2}{2} - b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] \\
 &= \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 2b_1 - \frac{b_2}{2} \\ 0 & 0 & 1 & 2 & \frac{b_2}{2} - b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] = [R \ d]
 \end{aligned}$$

$$Ax = b \Leftrightarrow Rx = d$$

Thus $Ax = b$ is solvable if $b_3 - b_2 + 4b_1 = 0$

(b) Find the complete solution to $Ax = b$ with $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

$$b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \Rightarrow d = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

$$Rx = d \Rightarrow \begin{cases} x_1 + 3x_2 = \frac{1}{2} \\ x_3 + 2x_4 = \frac{1}{2} \end{cases}$$

let $x_2 = x_4 = 0$, we get

$$x_p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

$$Rx = 0 \Rightarrow \begin{cases} x_1 + 3x_2 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$x_2 = 1, x_4 = 0 \Rightarrow s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 = 0, x_4 = 1 \Rightarrow s_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\text{Thus } x = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} - 3C_1 \\ C_1 \\ \frac{1}{2} - 2C_2 \\ C_2 \end{bmatrix}$$